Section A: Multiple choice questions

There are 20 questions in this section. Please answer all questions at the start of your answer paper. Each question is work 1.5 marks.

1) One important consequence of a good being scarce is that:
   a) There will be a situation of excess demand
   b) There will be a situation of excess supply
   c) The price of the good will be greater than zero
   d) There will be no equilibrium in the market for the good

2) Since travel by train and air are substitutes, we expect that a reduction in train fares will lead to
   a) An increase in the demand for air travel
   b) A decrease in the demand for air travel
   c) An increase in the supply of air travel
   d) A decrease in the supply of air travel

3) Due to the increase in the number of cancer cases, the government has decided to increase the specific tax on cigarette consumption in order to deter smoking. The demand for cigarettes is relatively price inelastic due to the presence of habitual smokers. Which of the following statements will be true about the impact of the specific tax?
   a) Consumers pay more taxes than producers
   b) Producers pay more taxes than consumers
   c) Consumers and producers pay taxes by equal proportions
   d) Consumers pay all taxes

4) If there is a price floor of £40 set for the course textbook, and the current market price is £45, then, because of the price floor
   a) There will be excess demand for the textbook
   b) There will be excess supply of the textbook
   c) The market price will fall to £40
   d) There will be no change in the market equilibrium

5) If a good is abundant then we expect that
   a) The market price will be zero, and there will be excess supply
   b) The market price will be zero, and there will be excess demand
   c) The market price will be greater than zero, and the market will clear
   d) The market price will be less than zero, and the market will clear

6) When increasing amounts of a variable factor are used with a given amount of fixed factor, there will come a point when each extra unit of the variable factor will produce less additional output than the previous unit. This is known as:
   a) Opportunity cost
   b) Diseconomies of scale
   c) Diminishing marginal cost
   d) Diminishing marginal returns

7) In the short-run, a firm is making losses. If the firm continues production, it will earn revenue of £400 but incur variable cost of £300. The fixed cost is equal to £200. What should the firm do?
   a) Continue production, since losses are relatively small
   b) Shut down production, since the firm cannot make a profit
   c) Shut down production, since revenue cannot be increased
   d) Continue production, since total revenue exceeds total variable cost

8) In the very short (immediate) run, which of these alternatives is available to firms in response to an increase in demand for their product?
   a) inform customers they will have to wait
   b) start proceedings to build a new factory
   c) begin a programme of Research & Development
   d) introduce overtime

9) A firm could guarantee increased profits by:
   a) increasing the price for elastic products
   b) using advertising to boost sales
   c) achieving product differentiation through expenditure on research and development
   d) maintaining the output and the price it charges, but reducing costs

10) Which of these situations do we associate with the model of monopolistic competition, rather than analysis of firms in oligopoly?
   a) Firms producing a virtually identical product, differentiated by branding
   b) A few firms sharing a large proportion of industry output
   c) Various barriers to the entry of new firms
   d) Firms taking account of the behaviour of competitors in their planning

11) Which of the following is the most likely to cause increasing inflationary pressures in an economy?
   a) a rise in the nation's exchange rate
   b) a rise in the top rate of income tax from 40% to 45%
   c) a fall in the level of unemployment
   d) A reduction in the rate of VAT.

12) Which of the following is an example of expansionary fiscal policy?
   a) A cut in interest rates.
   b) An increase in money supply.
   c) Labour market reforms making it easier to hire and fire workers
   d) A cut in the basic rate of income tax

13) You are given the following information on a simple closed economy:
   Autonomous consumption expenditure: 10 million;
   Marginal propensity to consume: 0.75;
   Investment expenditure: 20 million ;
   Government expenditure: 40 million;
   Net Taxes are zero.
   The equilibrium level of national income will be:
   a) 280

b) 140
c) 105
d) 70

14) A country is said to be moving into a recession when:
a) actual output falls below the full employment level of output.
b) actual output falls below the trend level of output.
c) actual output falls.
d) actual output falls for two consecutive quarters

15) Which of the following arguments made in favour of trade barriers offers potential benefits to consumers?
a) Trade barriers will soften the fall of declining industries.
b) Trade barriers will protect industries from competition from low wage countries.
c) Trade barriers will protect industries from dumping.
d) Trade barriers will nurture infant industries by protecting them from cheaper imports

16) Which of the following statements counters the claim that governments can always increase national output by increasing its expenditure?
a) The marginal propensity to consume is always less than one.
b) The labour force is not a fixed percentage of the population
c) There is an upper limit to national output determined by the economy's resources.
d) The ratio of capital to output increases as national output increases.

17) Which of the following policies would be appropriate for dealing with a deflationary gap?
a) Increase income taxes.
b) Increase subsidies to exporting industries.
c) Decrease unemployment compensation.
d) Decrease government expenditure.

18) The following data refer to a hypothetical economy:

Year	**Potential GDP (full employment)**	**Employment index**	**Price index**
2020	100	100	100
2021	103	98	102

Comparing the data for 2020, and 2021:
a) GDP per capita increased by 3%.
b) There was inflation.
c) Real GDP decreased.
d) Money GDP increased or was constant.

19). A rise in interest rates is good for the economy as saving would increase and funds for investment would rise creating growth. This statement is incorrect because
a) Savings and investment are not related

b) Growth is not related to capital accumulation
c) More saving means less consumption
d) More saving means higher inflation

20) In an economy faced with high inflation and a falling exchange rate, what would be the most appropriate monetary policy to adopt?
a) Reduce interest rates
b) Increase the budget deficit
c) Reduce the money supply
d) Decrease income tax

Section B: Short answer questions

There are four questions in this section, each worth 10 marks. You should answer all four questions.

1) In the UK, the government has anticipated a rapid increase in the price of energy this winter by intervening in the market.
   a) What do we measure using the price elasticity of demand, and the price elasticity of supply? [2 marks]
   PED is the responsiveness of the quantity demanded to a change in market price; PES is the responsiveness of quantity supplied to a change in the market price.
   b) State three factors which are likely to determine the price elasticity of demand for energy among UK households. [3 marks]
   Price and availability of substitute goods; importance of good in total consumption; time in making choices, allowing diversification
   c) The government's intervention will reduce the market price of energy to consumers by up to 60%. Drawing on the three factors stated in part b), explain briefly why we might expect household demand for energy to be price inelastic across the winter. How will your answer differ for low, and high, income households. [5 marks]
   Households need energy in winter for warmth. There will be few substitutes, especially give the short time horizon, for households to change their source of energy for heating. For lower income households, energy costs will be a substantial fraction of their budget. We expect household demand for energy to be price elastic this winter, with the price elasticity higher for low income than other households.

2) A steel producer has high fixed costs of 7200 per week. We also have some information about its total costs of production.

Output	0	100	200	300	400	500	600	700
TC	7200	7950	8400	8600	8800	9250	12250	16000

   a) For each of the output levels, calculate the steel producer's average variable costs (AVC), average (total) cost (AC) and the marginal cost (MC). [4 marks]

Output	0	100	200	300	400	500	600	700
MC								
AC	-							
AVC	-							

Output	*0*	*100*	*200*	*300*	*400*	*500*	*600*	*700*
MC		*7.5*	*4.5*	*2.0*	*2.0*	*4.5*	*30.0*	*37.5*
AC	*-*	*79.5*	*42*	*28.67*	*22*	*18.5*	*20.42*	*22.86*
AVC	*-*	*7.5*	*6*	*4.67*	*4*	*4.1*	*8.42*	*12.57*

b) Sketch a diagram showing the marginal cost curve, the average variable cost curve and the firm's short run supply curve to the diagram. [3 marks]

The MC and AVC curves should be drawn using the figures calculated in 2b).

The firm's short-run supply curve is the portion of its marginal cost curve that lies above its average variable cost curve.

c) Explain the relationship between high fixed costs and a firm's economic vulnerability. [3 marks]

A typical firm will have a U-shaped AC curve.

If a firm has a high ratio of fixed factors to variable factors, then it is likely to face a steeper AC curve at lower levels of output (as output falls, high FC are spread over fewer and fewer units).

If a firm has a steeper AC curve than normal, and wishes to cut back on production (perhaps due to a recession), its AC rise much more quickly than the AC of a firm with a flatter AC curve.

The steeper its AC curve, the more vulnerable a firm will be (as any change in output has a greater effect on its costs and hence on its profit).

3) In an economy where taxes equal £40m, exports, government expenditure and investment expenditure are each equal to £200m. Households consume 75% of their disposable income. Imports equal 25% of national income.

a) Define the concept of the circular flow of income in the economy, illustrating your answer. [4 marks]

Diagram with flow of goods and services from firms to households, and supply or factors of production from households to firms, with financial flows (payments for goods and services, wages of labour + interest on capital) flowing in the opposite direction.

b) For this economy, calculate the equilibrium level of consumption expenditure.

First calculate the equilibrium level of national income.

For equilibrium, $Y \equiv C + I + G + X - M$

Consumption expenditure equals 75% of net income, i.e. $C = 0.75(Y - T) = 0.75\ Y - 30$

So $Y = 0.75\ Y - 30 + 200 + 200 + 200 - 0.25\ Y$

Then $Y = 0.50\ Y + 570$, and $Y = 1140$

From the consumption function, $C = 0\ .75(1140 - 40) = 825$

4) Two Countries, A and B, have the following production data for Bicycles and Apples. Each country has 100 resources and devotes half of these resources (50 resources) to each. Total production is 15,000 Bicycles and 175,000 tonnes of Apples.

	Bicycles ('000s)	**Apples ('000 tonnes)**
Country A	10	100
Country B	5	75
TOTAL	**15**	**175**

a) Which country has an absolute advantage in Bicycles? [2 marks]
Country A can make twice as many as Country B from the same resources, the advantage is 10:5 or 2:1.

b) Explain which country has a comparative advantage in Apples. [3 marks]
Domestic opportunity cost ratios.
For Country A the 'cost' of 1 Bicycle is 1000/100 = 10 tonnes of Apples.
For Country B the cost of 1 Bicycle is 750/50 = 15 tonnes of Apples.
For Country A the 'cost' of 1 ton of Apples = 10/100 = one-tenth of a Bicycle.
For Country B the 'cost' of 1 ton of Apples = 5/75 = one-fifteenth of a Bicycle.
In terms of the opportunity cost (the amount of Bicycles given up for one ton of Apples), Apples are lower opportunity cost in Country B than in Country A.
*Country B therefore has a **comparative advantage** in Apples.*

c) Country A and Country B may trade based on this comparative advantage, with the world terms of trade being that 1,000 bicycles trade for 12,500 tonnes of apples. Obtain the quantities produced, and the levels of consumption of both goods, given that each country specialises in the production of the good for which it has a comparative advantage, and that there are 8,000 bicycles exported. [5 marks]
With specialisation, country A produces 20,000 Bicycles, and country B will produce 150,000 tonnes of Apples.
With the world terms of trade, A and B can exchange 8,000 bicycles for 100,000 tonnes of apples; this leaves A with 12,000 bicycles, and 100,000 tonnes of apples, and B with 8,000 bicycles and 50,000 apples

Section C – Discussion questions
There are five questions in this section, each of which is worth fifteen marks. You should answer any two questions.

1) In some major East Asian cities, large 'megamalls' are emerging quickly and taking over the role previously fulfilled by smaller, family-owned shopping centres. This shift, from a large number of small shopping centres, to a small number of large megamalls, has resulted in a major change in the shopping experience.
Write a short report setting out

a) The economic advantages which large malls have over smaller shopping centres;
b) How property companies, which own large megamalls, earn revenue; and
c) The nature of the negative externalities associated with traffic congestion and engine emissions around these malls, explaining why malls are unlikely to ensure a socially optimal outcome.

Larger malls can experience economies of scale, an advantage that the smaller shopping centres are unable to enjoy. Moreover, there may be more economies of scope for megamalls too, since their size enables them to accommodate a greater product variety.
The small number of large megamalls also implies a more concentrated market structure, thus enabling the megamalls to enjoy more market power and thus retain a larger proportion of supernormal profits.
Revenue could be generated through rental income – by charging tenants who might rent space to carry out their respective business activities within the megamall. As the number of tenants are likely to be substantial, the rental rate need not be high as the business owners could capitalise on scale/volume of tenants. However, since there are not that many megamalls (market is concentrated), tenants may not have that many options to ply their trade, so the demand for rental space in the megamalls are likely to be price inelastic, enabling mall owners to raise rentals sharply.
Another source of revenue could be from parking fees charged to customers to visit the malls. Price discrimination could be practiced by charging different rates for different days.
The negative externality is an external cost imposed on a third party. This cost is not internalised by the market participants, so no one takes responsibility for it. This situation can be depicted using a simple diagram, with private marginal cost, social marginal cost and social marginal benefit schedules. It can be demonstrated that businesses tend to 'overproduce' leading to too much production than is optimal.

2) Explaining how prices are determined by the interaction of demand and supply, discuss whether record wheat and lamb prices in the last year may have led to farmers making record profits.

Prices are pushed up due to an increase in demand or a reduction in supply. This can be illustrated on a diagram.
Demand could have increased because people had more bbqs due to Covid restrictions, etc.
Supply could have fallen because of increases in the costs of production or shortages of factors of production.
Reference increased costs of production – students could mention the increase in fuel and energy costs, steep increases in fertiliser prices, prices of sheep feed increasing (because wheat and barley prices have increased). They could talk about the rate of inflation and how input prices have gone up and well as the prices of lamb and wheat.
It is worth mentioning supply shortages due to the war in Ukraine and shortages of farm workers (due to Brexit) and a lack of lorry drivers to transport goods.

Need to understand that total profits are determined by the difference between total costs and total revenue.
Students could use a diagram to explain how the gap between the total revenue and total cost curves shows profit level. A total profit curve can be added to the diagram.
Students could also talk about determining the profit maximising price and output using marginal costs and marginal revenue. This can be taken a step further by discussing how total profit can be calculated by using the average revenue and average cost curves.

3) Outline the elements of a PEST analysis, setting out considerations which you would expect to see when applied to a motor vehicle manufacturer, such as Volkswagen.
Political; Economic; Social; Technological
A wide range of material can be introduced, but discussion based in the class is likely to include the following.
Probably the largest long-term challenge facing vehicle manufacturers at the moment are two problems for which there will be technological solutions: switching to electric motors, as part of the political imperative to reach net zero, and the development of AI systems, enabling autonomous vehicles. Satisfactory answers will comment on associated political factors – government policy guiding the transition to electric vehicles, and licencing autonomous cars. Good answers will touch on social issues such as campaigns to reopen cities to pedestrians, changing the relationships with vehicles – and with autonomous vehicles, there may also be incentives to move from ownership or leasing of vehicles to short term rental – a long term aspiration for ride sharing firms such as Uber. Economic considerations can be largely macroeconomic, since a new car is a valuable asset, and manufacturers now make more money from providing finance than from manufacturing. However on the microeconomic side, there are also many barriers to entry in the industry, and so we expect firms to be able to exploit market power.

4) Evaluate the advantages and disadvantages of trading globally.

Advantages
- *Specialisation in production*
- *Absolute and comparative advantage.*
- *Lower prices by specialising in goods with lower opportunity cost.*
- *Injection into circular flow of income and increased GDP.*
- *The creation of more jobs.*
- *Increases in consumer income.*

Disadvantages
- *Firms move production to countries with less protection for workers and environment.*

- *Cultural issues*
- ***Dumping.***

Foreign monopolies
Essentials – e.g. defence.
Trade gap between manufactured and primary products.

5) "It is not the government's tax cuts which will do most to stimulate the economy; it is the subsequent multiplier effects which will create the most jobs." Explain what you understand "multiplier effects" to be, and how they work. In your discussion, set out factors in the real world which might limit multiplier effects

A tax cut (T) will increase households' disposable incomes (Y), i.e. (Y - T).
Part of this increase will be spent thereby increasing AD and the incomes of the suppliers of these goods and services.
Those suppliers will in turn spend part of this increase and so the multiplier process evolves with the increasing rounds of additional expenditure.
The proportion of additional income spent – the marginal propensity to consume (MPC) - determines the value of the multiplier.
Factors affecting the multiplier might include:

- *Switch from domestically produced to foreign produced goods*
- *Households start to save a higher proportion of additional income*
- *An increase in interest rates*
- *If the economy is close to full employment a potential output constraint could limit the multiplier effects.*
- *MPC and MPI*
- *Accelerator*
- *Exogenous shocks*

--- END OF EXAMINATION PAPER ---